

Deep Learning Basics

Session Agenda

Duration: 3 Hours

Session Overview

This session introduces the foundational concepts of deep learning, progressing from biological inspiration to practical neural network architectures. Learners will engage with concept-building lectures, and demos.

Learning Outcomes

- Understand the motivation and intuition behind deep learning
- Describe the structure and function of artificial neural networks
- Explain key training concepts: loss functions, backpropagation, optimizers
- Identify and apply common activation functions and regularization techniques
- Build and train a simple neural network using a modern framework
- Recognise popular architectures: CNNs, RNNs, and LSTM

Detailed Session Schedule

Duration	Topic / Content
Part 1 — Foundations of Neural Networks (85 min)	
10 min	Introduction to Deep Learning Objectives, structure, motivation and intuition behind deep learning
15 min	Biological Neuron vs Artificial Neuron Inspiration from the brain, perceptron model, weights and bias
20 min	Artificial Neural Networks (ANN) & Forward Propagation Layers, depth, width, and forward pass through the network
20 min	Activation Functions & Loss Functions Sigmoid, ReLU, Tanh, Softmax; Mean Squared Error and Cross-Entropy loss
20 min	Demo I: Neural Network Learning in Action Forward propagation, loss calculation, backpropagation, and gradient descent — mathematics to Python implementation
10 min	Break
Part 2 — Types of Neural Networks (85 min)	
20 min	Artificial Neural Networks (ANN) & Convolutional Neural Networks (CNN) ANN as the foundation of deep learning for structured/tabular data; CNN architecture with filters, pooling layers, and feature maps designed for image recognition and computer vision tasks
20 min	Recurrent Neural Networks (RNN), LSTM Networks RNN for sequential and time-series data; LSTM as an advanced RNN handling long-term dependencies
10 min	Architecture Comparison & Choosing the Right Model Side-by-side comparison of ANN, CNN, RNN, and LSTM — understanding strengths, limitations, and industry use cases to select the right architecture for a given problem

Duration	Topic / Content
30 min	Demo II: Neural Networks in Action Live demonstration walking through an ANN on tabular data, a CNN and RNN on their respective data types, and a side-by-side comparison of ANN vs CNN on MNIST — showing how architecture choice shapes performance even on the same task.
5 min	Q&A, Resources & Next Steps Open floor for questions, key concepts summary